

Some guidelines :

1. These queries are based on the COMPANY DB. Now since the tables specially the table Employee is too big you can drop some columns for the sake of simplicity.
2. Fill up the tables and run the queries to get an output in each case not an empty set.
3. You can change the query like if u don't have an entry like 'Smith' change it to a surname which exists in your table.

Simple Nested Query using IN

1.Retrieve the names of employees who work on projects controlled by department 5.

```
SELECT Fname, Lname
FROM EMPLOYEE
WHERE Ssn IN (
    SELECT Essn
    FROM WORKS_ON
    WHERE Pno IN (
        SELECT Pnumber
        FROM PROJECT
        WHERE Dnum = 5
    )
);
```

2.Retrieve project names where an employee with last name 'Smith' works.

Scalar (Single-Value using >/<) Nested Query

3. Find employees whose salary is greater than the average salary of all employees.

```
SELECT Fname, Lname
FROM EMPLOYEE
WHERE Salary > (
    SELECT AVG(Salary)
    FROM EMPLOYEE
);
```

4. Find employees earning is less than the average salary of department 5.

Nested Query with ANY and ALL

5. Find employees earning more than all employees in department 4.

6. Find employees earning more than any employee in department.

Correlated Nested Query

7. Retrieve employees who have at least one dependent.

```
SELECT Fname, Lname
FROM EMPLOYEE E
WHERE EXISTS (
  SELECT *
  FROM DEPENDENT D
  WHERE D.Essn = E.Ssn
);
```

8. Retrieve employees who do NOT have dependents.

Double Nested Query (Two-Level)

9. Retrieve employees who work on all projects controlled by department 5.

Tuple Comparison Nested Query

10. Find employees who work the same (Project, Hours) combination as employee 'John Smith'.

Nested Query with COUNT (Aggregate inside)

11. Find employees having two or more dependents.

```
SELECT Fname, Lname
FROM EMPLOYEE E
WHERE (
  SELECT COUNT(*)
  FROM DEPENDENT D
  WHERE D.Essn = E.Ssn
) >= 2;
```

12. Find employees who work on more than 2 projects.

13. Find employees whose salary is the highest in their department.

Nested Query inside HAVING

14. Find departments where average salary is greater than the company average salary.

```
SELECT Dno
FROM EMPLOYEE
GROUP BY Dno
HAVING AVG(Salary) >
  (SELECT AVG(Salary) FROM EMPLOYEE);
```

15. Departments Whose Minimum Salary Is Greater Than Company Minimum Salary.

16. Departments Having More Employees Than Average Department Size

Basic LIKE with %

1. Find employees whose first name starts with 'J'.

```
SELECT Fname, Lname  
FROM EMPLOYEE  
WHERE Fname LIKE 'J%';
```

2. Find employees whose last name ends with 'son'.

3. Find employees whose address contains 'Houston'.

#Using _ (Single Character Match)

4. List employees whose first name has exactly 4 letters.

5. List the Employees whose first name starts with 'A' and ends with 'a'.

#Using NOT LIKE

6. Employees whose last name does NOT start with 'M'.

#Using LENGTH

7. List Employees whose first name length is greater than 6.

```
SELECT Fname  
FROM EMPLOYEE  
WHERE LENGTH(Fname) > 6;
```

8. List Employees whose last name length is less than 5.

#SUBSTRING Matching

9. Employees whose 3rd letter of first name is 'r'.

```
SELECT Fname  
FROM EMPLOYEE  
WHERE SUBSTRING(Fname FROM 3 FOR 1) = 'r';
```

- FROM 3 → start at 3rd position
- FOR 1 → take 1 character
- Checks if that character is 'r'

10. Employees whose 2nd letter of first name is 'a'.

